Mofe Omolade

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Education

University of British Columbia Vancouver, BC
Bachelor of Applied Science in Computer Engineering *Expected May 2029*
Relevant Coursework: Circuit Analysis, Computing Systems, Software Construction, C Programming

Experience

Electronics Engineering Intern May 2026 – August 2026
Fulcrum Air *Calgary, AB*

- Owned the complete design life-cycle of an 8-layer 4-channel motor controller PCB to drive 12V actuators at 3A
- Designed system architecture around Teensy 4.1 and IFX007T half bridges, supporting multichannel motor control, CAN communication, analog current sensing, and sub-millimeter Hall-effect positioning
- Implemented orthogonal routing, dedicated power/ground planes, and ground stitching with a 2.1 MHz switching node shield, maintaining PCB surface temperature rise under 25°C under continuous 150 W draw
- Engineered a power hold-up circuit and falling-edge ISR using a voltage divider and 1000µF bulk capacitor, enabling EEPROM backups of actuator positioning data within 5 ms of power disconnection
- Built a test bench and developed C++ firmware to validate PCB function, verifying MCU pinouts, voltage regulation, inrush current stabilization, CAN signal integrity, and bidirectional actuation under dynamic loads

Design Teams

Electrical Team Member September 2026 – Present
UBC Formula Electric *Vancouver, BC*

- Spearheading hardware design and system integration for high-voltage battery module daughterboards and thermistor sensing boards across the 600V tractive battery pack
- Transitioning cell balancing from internal ADBMS switching to discrete external MOSFETs, increasing balancing current to 200 mA to reduce cell equalizing time by 75%
- Designing communications architecture to integrate Maxim 1-Wire protocol with STM32 embedded platform

Projects

24V – 5V Buck Converter | *LTspice, Altium* August 2026 – Present

- Designed an asynchronous buck converter stepping down an 18V–25.2V input to 5V at up to 500 mA with 85% peak efficiency, maintaining Continuous Conduction Mode down to a 100 mA minimum load
- Developed an active Type III compensation network using an MCP6021 Op-amp to cancel the 1.57 kHz LC double pole, achieving a 61.8° phase margin and 22.3 dB gain margin at a 3.26 kHz crossover frequency
- Implemented an analog voltage-mode PWM controller at 100 kHz, driving an LM334 constant-current source into a TLC555 timer to generate a linear sawtooth ramp fed into an MCP6561 comparator
- Engineered RC soft-start circuit on the 2.5V reference to suppress overshoot and limit inrush current to 10 mA
- Routing a compact 4-layer PCB, isolating sensitive feedback and timing traces from the high di/dt switching node

Self-Balancing Robot | *C++, ESP32, I2C* July 2026 – August 2026

- Developed C++ drivers for the MPU-6050 IMU sensor to parse gyroscope and accelerometer data over I2C
- Implemented non-blocking sensor data acquisition and applied a complementary filter to fuse accelerometer and gyroscope readings, achieving low-latency angle estimation while reducing gyroscope angular drift by 85%
- Engineering a custom closed-loop PID controller from first principles without third-party frameworks, incorporating anti-windup clamping, derivative filtering, and deadband compensation
- Programmed PWM generation using ESP32-C3 hardware timers to drive an L298N dual-motor bridges

Technical Skills

Tools: Altium, LTspice, STM32CubeIDE, MATLAB, Quartus Prime, Questa, Git
Hardware: CAN, SPI, I2C, UART, RS-485, STM32, ESP32
Languages: C/C++, Java, SystemVerilog
Lab: Oscilloscope, Logic Analyzer, Digital Multimeter, Function Generator, Soldering